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Correlation between threshold voltage and channel dopant concentration in negative-type metal-oxide-semiconductor field-effect transistors studied by atom probe tomography

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The correlation between threshold voltage (V_T) and channel boron concentration in silicon-based 65 nm node negative-type metal-oxide-semiconductor field-effect transistors was studied by atom probe tomography (APT). V_T values were determined for one million transistors in a single chip, and transistors having a $\pm 4\sigma$ deviation from the median V_T were analyzed using APT. V_T and the channel boron concentration were positively correlated. This is consistent with the relationship between the average boron concentration of wafers implanted with different channel doses and the median V_T of the million transistors. APT is suitable for the study of dopant-distribution-based device failure mechanisms. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4730437>]

With the rapid progress in semiconductor manufacturing, silicon-based complementary metal-oxide-semiconductor field-effect transistors (MOSFETs) have been developed with high level of integration and scalability. The reduction of MOSFET feature sizes is one of the main goals in the manufacture of high-performance devices, but transistors will reach a scalability limit with the 8 nm node in 2019.¹

The characteristic variability of MOSFETs has become a significant obstacle, preventing a continuous decrease in their feature size.^{2–4} In particular, the threshold voltage (V_T) that determines the on/off current through the channel regions is an important electrical parameter. Variability in V_T is generally classified into two components: systematic and random. The systematic component originates from intra-wafer and intra-chip variability and has been reduced by process and configurational optimization.⁵ On the other hand, the random component exhibits no such positional correlation in wafers. The random component of V_T variability is dominant in present-day MOSFETs.^{6,7} Therefore, clarifying the origin of the random component is required to enable continued FET scaling. Several simulations based on electrical properties have predicted that the primary origin is a fluctuation of discrete dopant atom positions in the channel region;^{2,8–11} however, it was impossible to confirm this prediction due to the lack of a suitable experimental evaluation method.

It was also reported that the random component in a negative (n)-type MOSFET is larger than in a positive (p)-type MOSFET.^{12,13} Previous atom probe studies indicated that the source/drain extension formation enhanced channel dopant variability, especially in n -type MOSFETs.¹⁴

However, the correlation between V_T and channel dopant concentration for individual transistors was not determined. Laser-assisted atom probe tomography (APT) is the only metrology capable of obtaining a three-dimensional elemental distribution with nearly atomic-scale resolution in such local regions and has been shown to be useful for distribution analysis in device structures.^{14–21} To understand the origin of V_T variability, direct observation of the channel dopant concentration in an actual MOSFET whose V_T has been identified is required. For further clarification, the observation of transistors with greatly differing V_T values is required.

In this work, we directly obtained the channel dopant concentration in actual n -MOSFETs whose V_T ranged widely from -3.9σ to $+4.0\sigma$ deviation from the median V_T (V_{Tm}) and studied the correlation between V_T and channel boron concentration. This deviation corresponds to less than one in a thousand transistors; therefore, suppression of specimen fracture during the measurement was necessary. We optimized the needle fabrication conditions, including needle shape and positioning of the target area, and greatly reduced the probability of specimen fracture during the APT measurements.

Our samples were actual 65 nm node n -MOSFETs, with a gate length and width of 60 and 140 nm, respectively. For controlling V_T , we used a conventional ion implantation technique. $^{11}\text{B}^+$ ions were implanted with a dose of $2 \times 10^{13} \text{ cm}^{-2}$ at 8 keV into Si(100) substrate as a channel. Subsequently, a 2-nm-thick gate oxide film and a polycrystalline silicon gate were formed, and a P^+ dose of $5 \times 10^{15} \text{ cm}^{-2}$ at 10 keV was implanted into the gate. Next, the sample wafer was annealed at 900 °C for 10 s. After the gate patterning, an As^+ dose of $1.2 \times 10^{15} \text{ cm}^{-2}$ at 3.5 keV was implanted as a source/drain extension, and activation annealing was performed at 900 °C for 30 s. And B^+ dose of

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$4.4 \times 10^{13} \text{ cm}^{-2}$ at 8 keV was implanted as a halo. Finally, rapid thermal annealing of the wafer was used to activate the dopant. For V_T measurement, we used a device matrix alloy test elemental group, which includes one million (1024×1024) MOSFETs in a chip. The V_T of about one million MOSFETs in a chip was measured using the Agilent N9201A Array Structure Parametric Test System in the linear region (drain bias: 50 mV). The detail of measurement procedure was described in Ref. 7.

Specimens for APT were lifted from the wafer using a gallium focused ion beam.^{22,23} Then, the specimens were sharpened into a needle shape using annular milling patterns. The apex of needle-shaped specimens was fabricated in the center of the gate. The sidewalls and shallow trench isolation were completely eliminated, as indicated by the broken curve in Fig. 1(a), since the presence of region—which has high threshold evaporation field in a specimen—leads to a frequent fracture during APT measurements. We increased the successful yield of measurement from 20% to over 80% by performing processing not to include STI in a needle-shaped specimen and it enabled us to perform the APT measurement of rare transistors in this work. We employed a laser-assisted local electrode atom probe (LEAP3000X HR, Cameca, AMETEK). The apparatus had an ion detection efficiency of about 37%. The specimens were cooled to 50 K during the APT measurement, and field evaporation was induced by a combination of a high standing voltage and a pulsed laser (wavelength: 532 nm, pulse energy: 0.5 nJ) with a frequency of 200 kHz. The ratio of doubly to singly charge state for silicon ($\text{Si}^{++}/\text{Si}^+$) in channel region was ranging from 30 to 250 in this work.

Figure 1(b) shows a normal probability plot of the V_T of one million n -MOSFETs in a single chip, and the frequency distribution is shown in the inset. The distribution of the data appears linear, indicating that the V_T variability obeys a normal distribution. We obtained APT measurement data from 21 specified transistors whose V_T ranged from 0.37 V to 0.71 V, corresponding to -3.9σ to $+4.0\sigma$ deviation from V_{Tm} ($\sigma \sim 0.042$ V), as shown by the large open squares in Fig. 1(b). An example of a 3D elemental map of an actual

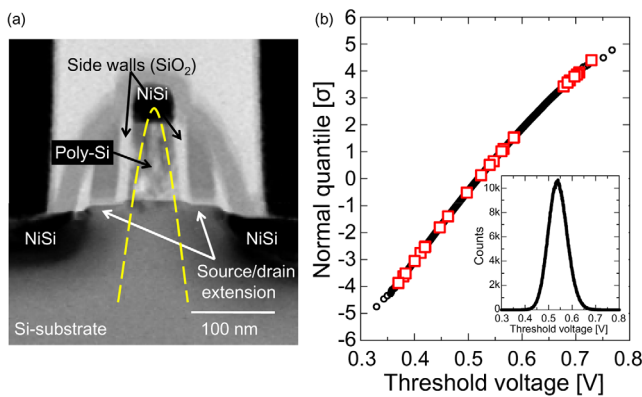


FIG. 1. (a) Cross-sectional transmission electron microscopic image of a typical 65 nm-node MOSFET. The broken curve shows the position of the needle specimen examined by APT. (b) Normal probability plot as a function of V_T of one million n -MOSFETs. The V_T measured using the Agilent N9201A Array Structure Parametric Test System are plotted as open circles. The transistors analyzed by APT in this work are plotted as open squares. The frequency distribution of the V_T is shown in the inset.

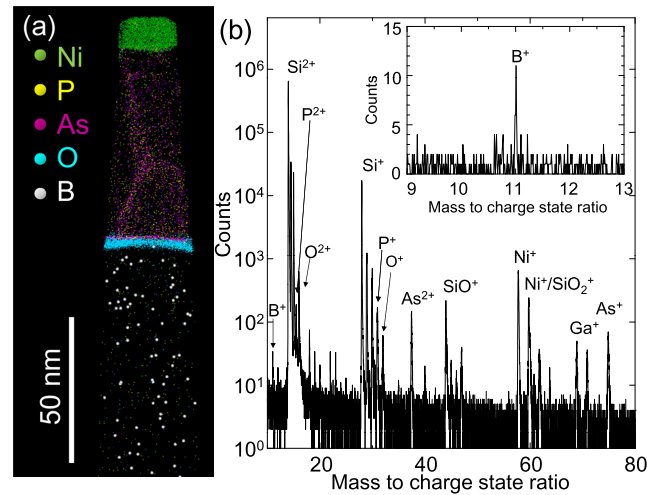


FIG. 2. (a) Three-dimensional elemental maps of actual 65 nm node n -MOSFET ($V_T = 0.56$ V). Nickel, phosphorus, arsenic, oxygen, and boron atoms are shown. Silicon atoms are not shown. (b) A representative mass spectrum of an actual n -MOSFET obtained by APT. An enlarged mass spectrum around the $^{11}\text{B}^+$ signal (mass/charge) in the channel region is shown in the inset.

n -MOSFET ($V_T = 0.56$ V) is shown in Fig. 2(a). To more clearly show the dilute dopant atoms, the matrix silicon atoms were not plotted. The principal dopant (boron, phosphorus, and arsenic), nickel, and oxygen atoms in the n -MOSFET were observed. The phosphorus and arsenic atoms in the gate were segregated at the grain boundaries of the polycrystalline silicon and the interface between the gate and gate oxide. Similar phenomena were previously observed in laterally uniform structures.²⁴ In this work, we focused on the channel dopant concentration in the 30 nm depth region just below the gate-oxide. This is considered to be an inversion layer and is a critical factor determining V_T of the MOSFET in the linear region. A representative mass spectrum of the entire measured region is shown in Fig. 2(b). The enlarged mass spectrum around the $^{11}\text{B}^+$ signal in the analyzed channel region (the $25 \times 25 \times 30 \text{ nm}^3$ box just below the gate oxide) is shown in the inset. The boron peak to background noise ratio was over 10:1. This level is low enough to evaluate the dilute ($3 \times 10^{18} \text{ atom} \cdot \text{cm}^{-3}$) channel boron concentration.

Figure 3 shows the correlation between V_T and channel boron concentration in the same volume just below the gate oxide. The standard deviation (uncertainty) of the real boron concentrations estimated from the number of the boron atoms detected with 37% efficiency were used as the error bars in Fig. 3. The fitting line of the APT data is shown as a solid line in Fig. 3. In the 30 nm depth just below the gate, V_T was positively correlated with the channel boron concentration, and the gradient of the fitting line was $(2.83 \pm 0.52) \times 10^{18} \text{ atoms} \cdot \text{cm}^{-3} \text{V}^{-1}$. For reference, we performed a similar analysis in the deeper region, but no significant correlation between V_T and the channel concentration was observed (the gradient of the fitting line was $(0.29 \pm 0.81) \times 10^{18} \text{ atoms} \cdot \text{cm}^{-3} \text{V}^{-1}$).

To further examine the correlation between V_T and the boron concentration variability in the channel inversion layer, we prepared transistors on five wafers that had been implanted with different doses of boron in the channel

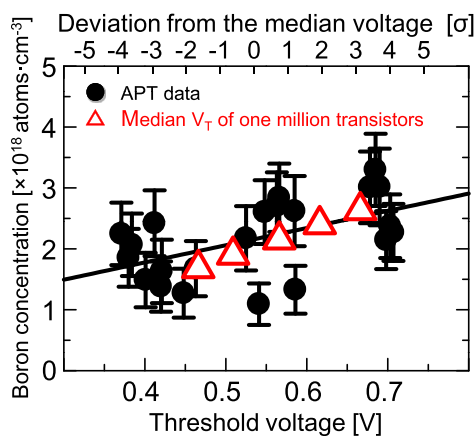


FIG. 3. Correlation between V_T and channel boron concentrations in the $25 \times 25 \times 30 \text{ nm}^3$ box just below the gate-oxide. Closed circles represent the APT data. Open triangles represent the estimated mean concentration of five different channel dose wafers that were measured by secondary ion mass spectrometry.

region. Subsequently, we measured the V_T values of one million transistors in each wafer and obtained V_{Tm} . The mean boron concentrations of the channel regions were estimated by secondary ion mass spectrometry in non-patterned regions on each wafer. The correlation between V_{Tm} and the channel boron concentrations of five wafers with different channel doses are plotted as large open triangles in Fig. 3. These observations were consistent with the correlation between V_T and channel boron concentration measured by APT.

Note that boron concentration variability was observed, even at similar V_T values. This variability may have originated from the limited observation volume due to the APT sample preparation method. The local channel dopants observed by APT in this work are not always the most important factor determining V_T . To more fully understand the origins of V_T variability, dopant distribution analysis of additional device regions is necessary, including the gate width direction and near the source/drain extension.

In summary, we performed APT analysis of channel dopants in actual devices whose V_T values were known. By suppressing specimen fracture during the APT measurements, three-dimensional elemental maps of very rare transistors were obtained. As a result, we found a positive correlation between threshold voltage and channel boron concentration in the 30-nm-deep region just below the gate oxide. These results demonstrate that APT is a powerful tool for investigating the origins of device failure and will likely have further applications in the failure analysis of electric devices.

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